



Pregnant women's lifestyles and exposure to endocrine-disrupting chemicals: A machine learning approach[☆]

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ABSTRACT

Women have ubiquitous exposure to various endocrine disrupting chemicals (EDCs) present in personal care products, food packaging, and processing. Pregnancy is a phase of increased vulnerability to environmental stressors. Therefore, we aimed to identify questionnaire based variables of pregnant women's lifestyle factors affecting the prenatal concentrations of EDCs: bis-phenol A (BPA), triclosan (TCS), parabens, and phthalates. We also aimed to explore the association between these lifestyle factors and EDC exposure in pregnant women in South Korea. This study is a part of Korean Children's ENvironmental health Study (Ko-CHENS). The following lifestyle factors: usage of personal care products, eating habits, cooking practices, food storage practices, and chemical exposure were evaluated through questionnaire. We examined prenatal EDCs: phenols (BPA), TCS, parabens (MEP, ETP, and PRP), and phthalates (MEHHP, MEOHP, MECPP, MBZP, MCOP, MCPP, MCNP, and MNBP). The random forest and least absolute shrinkage and selection operator regression machine learning models were used to predict the important lifestyle factors affecting the prenatal EDC concentrations in pregnant women. Next, we calculated the lifestyle score and evaluated its association with prenatal EDCs, respectively. Our results show that pregnant women who used makeup [β : 1.01, 95% C.I.: 0.01,2.00] >6 times/week had a significant increase in early-pregnancy (EP) Σ Paraben exposure. Using perfume up to 3 times/month was significantly associated with EP TCS exposure (β : 0.05, 95% C.I.: 0.01,0.23). While, using perfume >6 times/week was significantly associated to late-pregnancy (LP) Σ Paraben exposure, and consuming cup noodles significantly increased LP Σ DEHP exposure. Linear model analysis showed that the lifestyle score significantly increased the EP (β : 0.24, 95% C.I.: 0.07,0.40) and LP (β : 0.10, 95% C.I.: 0.01,0.20) Σ Paraben exposure. Therefore, pregnant women's lifestyle factors, such as using makeup and perfume and eating habits (e.g., cup noodle consumption), were associated with prenatal EDC exposure.

[☆] This paper has been recommended for acceptance by Dr Montes Marques.

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1. Introduction

Exposure to endocrine-disrupting chemicals (EDCs) has increased worldwide. EDCs are present in many consumer products, including personal care products, food packaging, and food storage items (Serrano et al., 2014; Woods et al., 2017). They dysregulate the endocrine system, affecting hormonal homeostasis. According to the World Health Organization (WHO), an endocrine disruptor is defined as “an exogenous substance or mixture that alters function(s) of the endocrine system and consequently causes adverse health effects in an intact organism, its progeny, or (sub) populations” (Ciallella et al., 2021). Studies have shown that women are more exposed to EDCs such as parabens, phthalates, and phenols than men because they are more likely to use lotions, cosmetics, perfumes, hair products, and soaps. Furthermore, phthalates and phenols are derived from food packaging and processing methods, leading to food product contamination (Haggerty et al., 2021; Braun et al., 2014).

The prenatal period can be a critical period for EDC exposure, causing adverse effects on birth outcomes, such as preterm birth, low birth weight, intrauterine growth retardation, small for gestational age, macrosomia, and large for gestational age; consequently, they negatively affect maternal and child health (Haggerty et al., 2021). Prenatal EDCs influence reproductive outcomes via inflammation, oxidative stress, hormonal and metabolomics changes, and changes in the microbiome and epigenetics (Haggerty et al., 2021). Most pregnant women in the United States and Canada have detectable EDCs in their blood or urine (Woodruff et al., 2011; Arbuckle et al., 2014). Industrial development and consumption behavior have also increased EDC exposure in Korean women.

Lifestyle factors are important and modifiable elements in pregnant

women. Although previous studies indicated the harmful effects of EDC exposure, pregnant women's awareness and knowledge are limited. As such, this study aimed to evaluate the modifiable lifestyle factors affecting the prenatal EDC exposure of pregnant women.

Exposure to substances involves multiple factors at a given time point and these multiple exposure sources of EDCs should be evaluated comprehensively. We used machine learning methods (MLMs) to predict the lifestyle factors affecting the prenatal EDC exposure of pregnant women. They (MLMs) are feasible approaches for exploring the prediction performance. The MLMs can process high-dimensional data and extract meaningful insights from diverse data with high accuracy (Chen et al., 2024). Their (MLMs) advantages include data-driven strategies and non-linearity and multi-collinearity modeling (Wu et al., 2022). The MLMs help identify exposure factors in complex data patterns and reveal potential interactions between different lifestyle factors. They (MLMs) overcome the inadequacies of robustness and overfitting of traditional regression methods (Oskar et al., 2021).

Our study aimed to predict the lifestyle factors affecting prenatal EDC exposure of pregnant women. We used a two-stage approach: (i) to identify questionnaire-based lifestyle variables, including the usage of personal care products, chemical exposure, eating habits, and cooking and food storage practices, which affected the exposure to prenatal EDCs (parabens, triclosan, phenols, and phthalates) via MLMs and (ii) to explore the association between the lifestyle factors and EDC exposure of pregnant women in South Korea.

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Figure 1. Selection of study participants

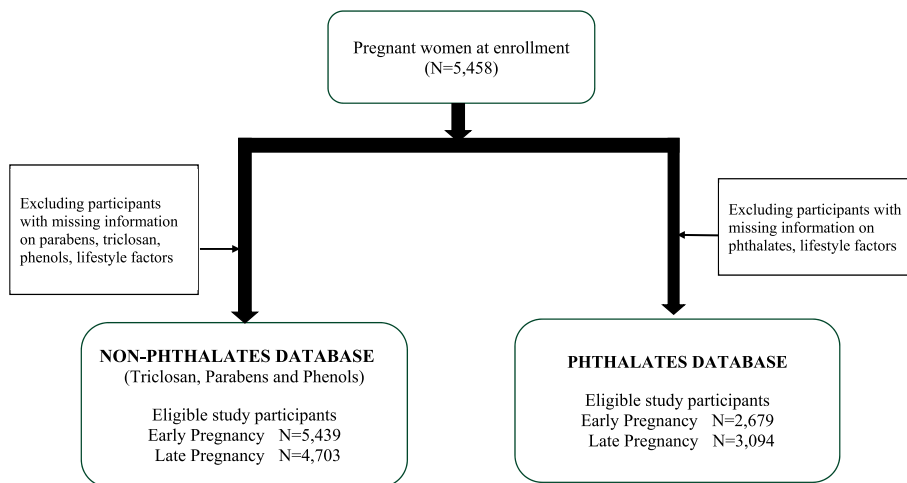


Fig. 1. Selection of study participants.

Table 1

The baseline characteristics of study participants at early pregnancy.

	Total study participants (N = 5458) (N[%]/Mean $\pm$ SD)	Non-phthalates database participants ^a (N = 5439) (N[%]/Mean $\pm$ SD)	Phthalates database participants ^b (N = 2679) (N[%]/Mean $\pm$ SD)
Age (years)	31.95 $\pm$ 3.8	31.9 $\pm$ 3.8	32.17 $\pm$ 3.87
Education			
High School	667[12]	666[12]	279 [10]
University	4791[88]	4776[88]	2401[90]
Income (10,000 KRW/month)			
<200	374[7]	373[7]	148[6]
200–400	2715[50]	2701[50]	1246[46]
>400	2369[43]	2365[43]	1285[48]
Parity			
1	3651[67]	3641[67]	1885[73]
>1	1562[33]	1559[33]	675[27]
Pre-pregnancy BMI (kg/m²)			
<18.5	554[11]	552[10]	259[10]
18.5–25	3829[73]	3819[74]	1891[74]
>25	830[16]	829[16]	410[16]

Numbers may not match due to missing values.

SD: Standard deviation.

^a Non-phthalate database include participants with data on phenols, triclosan and bis-phenol A.^b Phthalate database include participants with data on all phthalates. BMI: Body mass index.

2. Materials and methods

2.1. Study population

This investigation is a part of a nationwide birth cohort, Korean CHildren's ENvironmental health Study (Ko-CHENS), which has been described in detail in another study (Jeong et al., 2019). The core cohort in Ko-CHENS included 5458 pregnant women aged above 18 years in the early period of pregnancy (less than 20 weeks of gestational age) from 2015 to 2019. Our study excluded women with missing information on prenatal EDCs and lifestyle factors. Because more details on phthalates were missing, we made separate databases for non-phthalates and phthalates. The non-phthalate database included environmental phenolics (parabens and phenols such as bisphenol A [BPA] and triclosan [TCS]; $n = 5439$) at early pregnancy. The phthalate database covered only phthalates ($n = 2679$) at early pregnancy (Fig. 1). The

non-phthalate and phthalate databases comprised 4703 (86%) and 3094 (57%) study participants with late pregnancy (>28 gestational weeks), respectively (Fig. 1).

Written informed consent was obtained from all participants. This study was approved by the Institutional Review Board of Ewha Womans University Hospital (SEUMC, 2022-11-010). Upon enrollment, a detailed questionnaire was used to obtain the data on: demographic, socioeconomic, residential, medical, and reproductive history; exposure to occupational hazards; alcohol consumption; and nutritional habits.

2.2. Measurement of EDCs

Maternal urine samples were collected during early pregnancy (<20 gestational weeks) and late pregnancy (>28 gestational weeks), placed in polypropylene containers, processed, and stored at -80°C before analysis. They were analyzed for prenatal EDCs: (TCS, methyl-paraben [MEP], ethyl-paraben [ETP], propyl-paraben [PRP], BPA, mono(2-ethyl-5-hydroxyhexyl) phthalate [MEHHP], mono(2-ethyl-5-oxohexyl) phthalate [MEOHP], mono(2-ethyl-5-carboxypentyl) phthalate [MECPP], mono-benzyl phthalate [MBZP], mono-carboxyoctyl phthalate [MCOP], mono(3-carboxypropyl) phthalate [MCP], mono(carboxynonyl) phthalate [MCNP], and mono-n-butyl phthalate [MNBP]). The concentrations of phenols and phthalates were quantified via solid phase extraction and liquid extraction coupled with ultraperformance liquid chromatography–mass spectrometry, respectively. Next, electrospray ionization and tandem mass spectrometry was performed in accordance with previously published methods. The secondary oxidized metabolites, which constitute the majority of DEHP metabolites excreted in urine, were prioritized. Among all metabolites, as, mono-(2-ethylhexyl) phthalate (MEHP) represents less than 10% of the original DEHP dose and exhibits the shortest elimination half-life. Information on MEHP was not collected in our birth cohort (Kato K et al., 2004). Laboratory limits of detection (LOD) are listed in Table 2. EDCs below the LOD were assigned a value of the metabolite LOD/ $\sqrt{2}$ (Hornung and Reed, 1990). EDCs below the LOD of more than 50% were imputed using maximum likelihood estimation methods (Chen et al., 2011).

2.3. Lifestyle factor data

Data on lifestyle factors, including use of personal care products, chemical exposure, eating habits, and cooking and food storage practices, were collected through questionnaires distributed by the physician-in-charge at the time of participant enrollment. Such information described the participants' habits during the 3 months leading

Table 2

The distribution of prenatal endocrine disrupting chemicals in pregnant women.

EDCs ^a (μg/L)		LOD	% >LOD	Early pregnancy (N = 5439)		Late pregnancy (N = 4703)	
				Median	IQR	Median	IQR
ΣParaben ^c		–	–	2.30	2.10	2.20	2.10
	MEP	0.120	99	42.72	3.10	37.41	0.90
	ETP	0.129	98	34.38	2.90	41.04	0.80
	PRP	0.090	89	1.99	3.70	1.95	0.90
TCS		0.196	57	0.39	1.30	0.46	1.30
BPA		0.075	96	2.69	1.30	1.72	1.00
EDCs ^b (μg/L)		LOD	% >LOD	Early pregnancy (N = 2679)		Late pregnancy (N = 3094)	
				Median	IQR	Median	IQR
ΣPhthalates ^d		–	–	0.20	0.20	0.10	0.10
	MBZP	0.197	62	0.54	1.40	0.55	1.40
	MCOP	0.124	90	0.85	1.10	1.21	0.90
	MCPP	0.176	70	0.40	0.80	0.55	0.70
	MCNP	0.097	70	0.22	0.80	0.96	0.90
	MNBP	0.323	98	13.14	0.90	3.51	1.70
ΣDEHP ^e		–	–	0.10	0.10	0.10	0.10
	MEHHP	0.241	95	5.87	1.10	7.12	1.10
	MEOHP	0.100	99	4.81	1.00	5.97	0.90
	MECPP	0.231	100	10.15	0.90	4.84	1.70

Abbreviations: EDCs- Endocrine disrupting chemicals, IQR-interquartile range, LOD-Limit of detection, TCS -Triclosan, MEP- methyl-paraben, ETP- ethyl-paraben, PRP-propyl-paraben, MEHHP- mono (2-ethyl-5-hydroxyhexyl) phthalate, MEOHP- mono (2-ethyl-5-oxohexyl) phthalate, MECPP- mono (2-ethyl-5carboxypentyl) phthalate, MBZP-Mono-benzyl phthalate, MCOP-Mono-carboxyoctyl phthalate, MCPP-Mono(3-carboxypropyl) phthalate, MCNP-Mono (carboxynonyl) phthalate, MNBP-Mono-n-butyl phthalate, DEHP-diethylhexyl phthalate.

^a Non-phthalate database include participants with data on phenols, triclosan and bis-phenol A.

^b Phthalate database include participants with data on all phthalates.

^c Molar sum of ΣParaben=(MEP) +(ETP) +(PRP) expressed in μmol/L.

^d Molar sum of Phthalates= (ΣDEHP + (MCOP) +(MCPP) +(MBZP) +(MNBP) expressed in μmol/L.

^e Molar sum of ΣDEHP= (MEHHP) +(MEOHP) +(MECPP) expressed in μmol/L.

up to their enrollment in the study. Information on the use of personal care products included the participants' use of perfume, hair products, body wash, makeup, nail polish, and antibacterial products. Eating habits included the consumption of cup noodles, fast food such as hamburgers and pizzas, frozen food, popcorn, canned food, dairy products, and vinyl-packed drinks. Data on food storage included how frequently the participants stored hot food in a plastic container, hot water in a plastic container, ate delivery food, or used disposable cups. These data were categorized as follows: rarely eat, one to three times a month, one to three times a week, four to six times a week, one to three times a day, or four or more times a day. These frequencies were further categorized into no use, use up to three times per month, and use more than six times per week. Information on cooking practices such as using plastic microwave containers, plastic cooking tools, plastic teaspoons, plastic spoons, plastic dishware, and plastic storage containers was also obtained (categorized as yes or no). Chemical exposure was assessed by inquiring about the use of mosquito repellent, insecticides, pesticides, air fresheners, and disinfectants in the last 3 months (categorized as yes or no).

2.4. Statistical analysis

The participants and lifestyle factors were subjected to descriptive analyses by using the *t*-test for continuous variables and the chi-square test for categorical variables. Afterward, prenatal EDC concentrations were natural log-transformed and adjusted for creatinine because of their skewed distribution. The median and interquartile ranges of the prenatal EDC concentrations were calculated. As, exposure to many EDCs typically occurs at a given point in time, we used molar sum of Parabens, Phthalates and DEHP. Urinary parabens were summarized as molar sum measure: ΣParabens = (MEP/180.20) + (ETP/166.17) + (PRP/152.15). Urinary di(2-ethylhexyl) phthalate (DEHP) metabolites were quantified as a molar sum measure: ΣDEHP = (MEHHP/294.34) + (MEOHP/292.33) + (MECPP/308.33). The molar sum of all phthalates was expressed as follows: ΣPhthalates = ΣDEHP + (MCOP/322.36) +

(MCPP/252.22) + (MBZP/336.4) + (MNBP/222.24). The prenatal EDC concentrations were obtained separately at early and late pregnancy (Fruh et al., 2022).

2.4.1. Machine learning model

All machine learning models included data regarding the participants' lifestyle factors ($p = 28$), such as usage of personal care products ($p = 6$), cooking habits ($p = 6$), eating behaviors ($p = 7$), food storage practices ($p = 4$), and chemical exposure ($p = 5$). The prenatal EDC concentrations were examined for their normal distribution. Data were split into a training dataset (70%) and a validation dataset (30%). Machine learning models such as random forest (RF) and least absolute shrinkage and selection operator (LASSO) regression were used to predict the lifestyle factors affecting the exposure to ΣParabens, ΣPhthalates, ΣDEHP, BPA, and TCS individually at early- and late-stage pregnancy.

RF is a classification algorithm that generates multiple decision trees. It randomly selects variables and trains subsets when building each individual decision tree. The aggregated prediction results of each tree are used to create an uncorrelated forest of trees, whose predictions are more accurate than those of an individual tree (Wei et al., 2022). The RF model can identify nonlinear associations and interactions (Stafoggia et al., 2017). The variable importance score in the RF model is calculated based on the reduction of impurity or error attributable to each variable. The grid parameters for tuning the RF model include the number of variables to be considered for splitting at each node ($mtry$ was 1:10), and the minimum node size required for splitting ($min.node.size$ was 3). We used splitting rule as variance in the RF model. To apply the optimal hyper parameters, we conducted repeated 5-fold cross-validation to ensure robust model performance. In this study, the RF model was implemented using the "ranger" R package (Rocha et al., 2018).

LASSO regression is commonly used in a high-dimensional data setting for variable selection. It uses shrinkage through which the absolute values of regression coefficients are forced to be less than the tuning parameter lambda; thus, variable selection can be performed. We

Table 3A

Usage of lifestyle factors in pregnant women.

Personal Care Products, Eating Habits and Food Storage Practices	Not used N (%)	3 times/months N (%)	>6 times/week N (%)
Perfume	4009 (74)	820 (15)	661 (11)
Hair products	4398 (81)	348 (6)	693 (13)
Body wash	112 (2)	313 (6)	5014 (92)
Make up	539 (10)	571 (10)	4329 (80)
Nail color	4597 (85)	754 (13)	88 (2)
Anti-bacterial products	4040 (74)	517 (10)	882 (16)
Cup noodles	1835 (34)	1257 (23)	2347 (43)
Fast food	1211 (22)	3621 (67)	607 (11)
Frozen food	1807 (33)	3019 (56)	613 (11)
Microwave food	4201 (77)	631 (12)	607 (11)
Canned food	1006 (18)	3791 (70)	642 (12)
Diary food	176 (18)	546 (70)	4717 (12)
Vinyl pack drink	3795 (3)	803 (10)	841 (87)
Hot food in plastic	3189 (59)	1318 (24)	932 (17)
Hot water in plastic	4968 (92)	302 (5)	169 (3)
Plastic wrap delivery food	1392 (26)	3123 (57)	924 (17)
Disposable cups	2390 (44)	1378 (25)	1671 (31)

used repeated 5-fold cross-validation method to select the optimal value of lambda. Then selected the lasso model with the minimum value for lambda. The LASSO regression is suitably fitted for variable selection among highly correlated variables, thereby yielding good interpretability (Huang et al., 2020). In this study, it was executed using the packages “lasso” in R.

2.4.2. Model validation and variable importance

In the training data, each model was fine-tuned using the grid search algorithm, and its performance across 1000 cross-validated data sets (five folds and 200 repeats) was determined. These algorithms were then predicted on the validation dataset. The root mean square error (RMSE) and root mean absolute error (RMAE) were used to evaluate the best performance of the machine learning models. The RMSE was computed as the mean of the squared differences between the predicted and actual values; the RMAE was determined as the differences in the absolute mean (Qin et al., 2019). The best algorithm with the lowest RMSE and RMAE was selected to predict the important lifestyle factors affecting the exposure to prenatal Σ Parabens, Σ Phthalates, Σ DEHP, BPA, and TCS individually in early- and late-stage pregnancy. Multiple linear regression was also conducted to compare the predictive performances of the MLMs.

2.4.3. Multiple linear regression

The five-most important lifestyle factors at early and late pregnancy Σ Parabens, Σ Phthalates, Σ DEHP, BPA, and TCS exposure, respectively were summated to make a lifestyle score. In summation, we scored: no use as 0, usage up to 3 months as 1, and usage one to six times a week or more as 2. Then, they were added for the five important lifestyle factors to obtain a lifestyle score, which ranged from 0 to 10. The distribution of lifestyle scores is presented in Supplementary Table 1. The lower the lifestyle scores, the better. The early- and late-pregnancy exposures to Σ Parabens, Σ Phthalates, Σ DEHP, BPA, and TCS were natural log-transformed. Next, the association of the calculated lifestyle score with early- and late-pregnancy Σ Parabens, Σ Phthalates, Σ DEHP, BPA, and TCS exposure as dependent variables was determined by individual multivariable adjusted linear regression model. Each multivariable adjusted linear regression model was adjusted for maternal factors, including maternal age, pre-pregnancy body mass index (BMI: <18.5, 18.5–23, and ≥ 23 kg/m²), maternal educational level (High school or $\geq$ University), parity (0 or 1), gestational age (weeks), total calorie intake (kcal), and season of urine sample collection. Missing data for gestational age, education, pre-pregnancy BMI, and parity were included by assigning unknown category. The calculated lifestyle scores at early and late pregnancy were categorized according to their median value as low

and high groups, respectively. Then, linear regression analysis was performed to determine the association between lifestyle score groups and early- and late-pregnancy exposures to Σ Parabens, Σ Phthalates, Σ DEHP, BPA, and TCS as dependent variables, separately, adjusted for identical covariates. The association of the five important individual lifestyle factors selected from the best fit machine learning model with early- and late-pregnancy exposures to Σ Parabens, Σ Phthalates, Σ DEHP, BPA, and TCS was evaluated by individual linear model analysis.

Statistical significance was defined as a $p < 0.05$. Data were analyzed using SAS version 9.4 (SAS Institute Inc., Cary, NC, USA) and R software (version 4.3).

3. Results

The mean age at early pregnancy in the non-phthalate and phthalate databases was 32 years, and more than 85% of the women had an education level higher than the university level. Approximately 73% of the women were primiparous, and 74% of the pregnant women had a normal BMI (Table 1). In late pregnancy, approximately 89% of the women completed their education until the university level, and 72% were primiparous (Supplementary Table 2). Sensitivity analysis did not reveal significant differences between the participants in the phthalate database and those that were not included (Supplementary Table 3) possibly because this database was derived from the entire population and the study participants exhibited homogeneity.

The median early-pregnancy MEP, ETP, and PRP were 42.72, 34.38, and 1.99 μ g/L, respectively, while late-pregnancy MEP, ETP, and PRP were 37.41, 41.04, and 1.95 μ g/L, respectively. The early-pregnancy median values of MEHHP, MEOHP, MECPP, MNBP, MBzP, MCOP, MCPP, and MCNP were 5.87, 4.81, 10.15, 13.14, 0.54, 0.85, 0.40, and 0.22 μ g/L, respectively; the late-pregnancy MEHHP, MEOHP, MECPP, MNBP, MBzP, MCOP, MCPP, and MCNP median values were 7.12, 5.97, 4.84, 3.51, 0.55, 1.21, 0.55, and 0.96 μ g/L, respectively. The molar sums of paraben, DEHP metabolites, and phthalates were 2.3, 0.1, and 0.2 μ mol/L for early pregnancy and 2.2, 0.1, and 0.1 μ mol/L for late pregnancy, respectively (Table 2).

Table 3A shows the frequency of usage of personal care products, eating habits, and food storage practices of the study participants. Nearly 80% of the study participants used makeup, and 92% used body wash more than six times per week. Approximately 43% of the pregnant women consumed cup noodles, and 87% consumed vinyl pack products more than six times per week. The participants indicated the following consumption: 67% fast food, 56% frozen food, and 70% canned food three times per month. Plastic wrap delivery food was used by 57% of the pregnant women three times per month. In addition, 74% of the pregnant women used plastic storage containers, 71% used air fresheners, and 74% used disinfectants (Table 3B). Nearly 52% of the cohort used a mosquito repellent. The comparison of the lifestyle factors did not

Table 3B

Usage of lifestyle factors in pregnant women.

Chemical exposure and cooking practices.	No N(%)	Yes N(%)
Plastic container microwave cooking	3313 (61)	2126 (39)
Plastic cooking tools	3855 (71)	1584 (29)
Plastic teaspoons	5220 (95)	219 (5)
Plastic cooking spoons	5067 (93)	372 (7)
Plastic dishware	4764 (88)	675 (13)
Plastic storage container	1422 (26)	4017 (74)
Mosquito repellent	2594 (48)	2845 (52)
Insecticides	2745 (51)	2694 (49)
Pesticides	4748 (87)	691 (13)
Air freshener	1570 (29)	3869 (71)
Disinfectants	1369 (26)	4019 (74)

A: Frequency for use of personal care products, eating habits and food storage practices. B: Chemical exposure and cooking practices.

Table 4

Prediction performance of multiple linear regression and machine learning models.

EDCs (μg/L)	Early pregnancy			Late pregnancy		
	Linear model	Random forest	Lasso regression	Linear model	Random forest	Lasso regression
ΣParabens^b						
RMSE ^a	4.655	4.680	4.656	4.831	4.832	4.833
RMAE ^a	3.803	3.838	3.804	3.941	3.948	3.946
TCS						
RMSE ^a	1.384	1.386	1.382	1.200	1.206	1.203
RMAE ^a	0.981	0.984	0.978	0.833	0.834	0.835
BPA						
RMSE ^a	1.121	1.130	1.122	1.116	1.117	1.118
RMAE ^a	0.843	0.851	0.844	0.846	0.847	0.848
ΣPhthalates^c						
RMSE ^a	0.242	0.240	0.241	0.215	0.216	0.217
RMAE ^a	0.130	0.128	0.129	0.120	0.121	0.122
ΣDEHP^d						
RMSE ^a	0.091	0.089	0.091	0.054	0.056	0.055
RMAE ^a	0.066	0.064	0.066	0.035	0.037	0.036

Abbreviations: EDCs- Endocrine disrupting chemicals, TCS-Triclosan, BPA-bisphenol A, DEHP-diethylhexyl phthalate. MEP- methyl-paraben, ETP- ethyl-paraben, PRP-propyl-paraben, MEHHP- mono (2-ethyl-5-hydroxyhexyl) phthalate, MEOHP- mono (2-ethyl-5-oxohexyl) phthalate, MECPP- mono (2-ethyl-5-carboxypentyl) phthalate, MBZP-Mono-benzyl phthalate, MCOP-Mono-carboxyoctyl phthalate, MCPP-Mono(3-carboxypropyl) phthalate, MCNP-Mono (carboxynonyl) phthalate, MNBP-Mono-n-butyl phthalate.

^a RMSE: Root mean square error. RMAE: root mean absolute error. The smaller RMSE and RMAE the better machine learning model.

^b Molar sum of ΣParaben=(MEP) +(ETP) +(PRP) expressed in μmol/L.

^c Molar sum of ΣPhthalates= (ΣDEHP) +(MCOP) +(MCP) +(MBZP) +(MNBP) expressed in μmol/L.

^d Molar sum of ΣDEHP= (MEHHP) +(MEOHP) +(MECPP) expressed in μmol/L.

Table 5

Key life style factors influencing prenatal EDCs.

Ranking of the variable importance ^a	Early pregnancy EDCs				
	ΣParaben ^b	TCS	BPA	ΣPhthalates ^c	ΣDEHP ^d
1	Make Up	Perfume	Plastic storage container	Plastic wrap delivery food	Hot food in Plastic
2	Perfume	Plastic cooking tools	Air freshener	Make Up	Vinyl pack drinks
3	Plastic wrap delivery food	Frozen food	Frozen food	Fast food	Make up
4	Hair products	Insecticides	Mosquito repellent	Cup Noodles	Cup noodles
5	Disposable cups	Cup noodles	Canned food	Disposable cups	Plastic container microwave cooking

Ranking of the variable importance ^a	Late pregnancy EDCs				
	ΣParaben ^b	TCS	BPA	ΣPhthalates ^c	ΣDEHP ^d
1.	Makeup	Disinfectants	Pesticides	Plastic teaspoon	Frozen food
2.	Cup noodles	Vinyl pack drinks	Disinfectants	Frozen food	Plastic teaspoon
3.	Pesticides	Disposable cups	Plastic cooking tools	Hair products	Hair products
4.	Hot water in plastic	Pesticides	Plastic wrap delivery food	Perfume	Cup noodles
5.	Perfume	Frozen food	Disposable cups	Diary food	Diary food

Abbreviations: EDCs- Endocrine disrupting chemicals, MEP- methyl-paraben, ETP- ethyl-paraben, PRP-propyl-paraben, MEHHP- mono (2-ethyl-5-hydroxyhexyl) phthalate, MEOHP- mono (2-ethyl-5-oxohexyl) phthalate, MECPP- mono (2-ethyl-5-carboxypentyl) phthalate, MBZP-Mono-benzyl phthalate, MCOP-Mono-carboxyoctyl phthalate, MCPP-Mono(3-carboxypropyl) phthalate, MCNP-Mono (carboxynonyl) phthalate, MNBP-Mono-n-butyl phthalate, RMSE- Root mean square error, TCS-Triclosan, BPA-bisphenol A, DEHP-diethylhexyl phthalate[DEHP].

^a The important lifestyle factors affecting prenatal EDCs were selected from the best fit machine learning model having the less RMSE and RMAE value.

^b Molar sum of ΣParaben=(MEP) +(ETP) +(PRP) expressed in μmol/L.

^c Molar sum of Phthalates= (ΣDEHP) +(MCOP) +(MCP) +(MBZP) +(MNBP) expressed in μmol/L.

^d Molar sum of ΣDEHP= (MEHHP) +(MEOHP) +(MECPP) expressed in μmol/L.

show any significant differences between the included and excluded participants (Supplementary Table 4).

The coefficients of predictors in the final model of LASSO and linear regression analysis are described in Supplementary Tables 5 and 6. The number of predictors (features) in each final model for linear and LASSO regression are illustrated in Supplementary Table 7. We used “ranger” package for RF analysis due to its high performance, particularly in handling large datasets and providing faster computation times. However, due to limitation of ranger package we could not describe the selected predictors for final model in random forest (Supplementary

Table 7). The RMSE and RMAE indicated the goodness of fit of the machine learning models. The lowest RMSE and RMAE predicted in the validation dataset corresponded to the best model. The important life-style factors predicting prenatal ΣParabens, TCS, ΣDEHP metabolites, and ΣPhthalates were selected from the best-fit MLM (Table 4). The frequency of makeup use was an important lifestyle factor for prenatal ΣParabens, and perfume usage was a significant predictive factor for TCS. Placing hot food in a plastic container and consuming delivery foods were important factors for ΣDEHP and ΣPhthalate exposure in early pregnancy. Frozen food consumption and plastic spoon usage were

Table 6

Linear model analysis of association between lifestyle score and prenatal EDCs.

	β estimate and 95% confidence intervals				
	Σ Parabens ^a	TCS	BPA	Σ Phthalates ^b	Σ DEHP ^c
Early pregnancy					
Lifestyle score (Continuous)	0.24 (0.07, 0.40)*	0.02 (−0.02,0.07)	−0.01 (−0.02,0.02)	0.01 (−0.10,0.11)	0.03 (−0.02,0.08)
Lifestyle high score group (ref = low)	0.71 (0.12, 1.30)*	0.07 (−0.09, 0.22)	−0.02 (−0.09, 0.05)	−0.01 (−0.54,0.22)	0.08 (−0.12,0.28)
Late pregnancy					
Lifestyle score (Continuous)	0.10 (0.01,0.20)*	−0.01 (−0.03,0.01)	−0.01 (−0.04,0.01)	−0.29 (−1.34,0.74)	0.04 (−0.01,0.08)
Lifestyle high score group (ref = low)	0.31 (−0.05,0.66)	−0.01 (−0.08,0.06)	−0.01 (−0.08,0.06)	−1.91 (−5.47,1.65)	0.10 (−0.03,0.02)

All model was adjusted for mother's age, pre-pregnancy BMI, education, parity, gestational age, total calorie intake and season of urine collection. Lifestyle score was divided as high and low from the median value for respective EDCs. * $p < 0.05$.

Abbreviations: BMI-Body mass index, EDCs-Endocrine disrupting chemicals, TCS-Triclosan, BPA-bisphenol A, DEHP-diethylhexyl phthalate, MEP- methyl-paraben, ETP- ethyl-paraben, PRP-propyl-paraben, MEHHP- mono (2-ethyl-5-hydroxyhexyl) phthalate, MEOHP- mono (2-ethyl-5-oxohexyl) phthalate, MECPP- mono (2-ethyl-5-carboxypentyl) phthalate, MBZP-Mono-benzyl phthalate, MCOP-Mono-carboxyoctyl phthalate, MCPP-Mono(3-carboxypropyl) phthalate, MCNP-Mono (carboxynonyl) phthalate, MNBP-Mono-n-butyl phthalate.

^a Molar sum of Σ Paraben=(MEP) +(ETP) +(PRP) expressed in $\mu\text{mol/L}$.

^b Molar sum of Σ Phthalates= (Σ DEHP + (MCOP) +(MCPP) +(MBZP) +(MNBP) expressed in $\mu\text{mol/L}$.

^c Molar sum of Σ DEHP= (MEHHP) +(MEOHP) +(MECPP) expressed in $\mu\text{mol/L}$. The summation of paraben, DEHP and phthalates were natural log transformed.

important lifestyle factors for Σ DEHP and Σ Phthalate exposure in late pregnancy (Table 5).

Linear model analysis showed that the lifestyle scores significantly increased early-pregnancy (β estimate: 0.24, 95% C.I.: 0.07,0.40) and late-pregnancy (β estimate: 0.10, 95% C.I.: 0.01,0.20) Σ Paraben exposure (Table 6). The high-lifestyle-score group showed a significant increase in early-pregnancy (β estimate: 0.71, 95% C.I.: 0.12,1.30) Σ Paraben exposure (Table 6). Makeup use (β estimate: 1.01, 95% C.I.: 0.01,2.00) more than six times per week significantly increased early-pregnancy Σ Paraben exposure. Using perfume up to three times per month was significantly associated with early-pregnancy TCS exposure (β estimate: 0.05, 95% C.I.: 0.01,0.23) (Supplementary Table 8). Perfume usage of more than six times per week was significantly associated with late-pregnancy Σ Paraben exposure, and consuming cup noodles significantly increased late-pregnancy Σ DEHP exposure (Supplementary Table 9).

4. Discussion

Our study found that the frequent use of makeup and perfume was an important lifestyle factor predicting prenatal Σ Paraben and TCS exposure in pregnant women. The frequent consumption of cup noodles was also an important lifestyle factor predicting prenatal Σ DEHP exposure. Furthermore, a high lifestyle factor score was significantly associated with an increase in prenatal Σ Paraben exposure.

Using personal care products, such as cosmetics, perfume, and hair products, is increased among Korean women (Park et al., 2019; Han et al., 2020). This increase may be influenced by peer pressure and social media (Cho et al., 2017). Limiting the use of personal care products is associated to low levels of EDC exposure (Martina et al., 2012). In another study on 45 women of reproductive age in northern Italy, using plastic storage containers is associated with increased BPA concentrations (Di et al., 2021). Using hair oils and gels is related to increased phthalate exposure in pregnant women (Fruh et al., 2022). Ingle et al., found that using personal care products is associated with increased EDC concentrations in pregnant women (Ingle et al., 2019). While, a recent study on breastfeeding Korean women, consuming cup noodles and using consumer products such as air fresheners and lotions are linked to EDCs (Kim et al., 2020).

Prenatal EDC exposure can cause adverse birth outcomes such as low birth weight and preterm birth. EDCs may have long-term implications on maternal and child health (Goodrich et al., 2019). Prenatal EDC exposure may cause long-term metabolic changes and hormonal disruption, leading to adiposity, type 2 diabetes, and dyslipidemia in women. EDCs invoke inflammation and oxidative stress by modulating the immune system; they also elicit metabolomic changes by decreasing

metabolomic factors involved in glucose and lipid metabolism. They alter epigenetics and the microbiome by inducing the methylation of IGF2 and PPAR α (Padmanabhan et al., 2021; Montrose et al., 2018).

Our results emphasized that pregnant women's lifestyle factors mainly contribute to prenatal EDC exposure. We applied different machine learning methods: random forest, lasso regression, and linear model approaches, to determine the critical factors affecting EDC exposure. Luan et al. (2021) used a random forest model to explore the exposure of pregnant Chinese women to prenatal EDC. Similarly, Rocha et al. (2018) used a random forest algorithm to predict exposure to endocrine-disrupting chemicals in children. Epidemiological studies have used tree-based machine learning methods to examine and predict interactions between environmental pollutants (Midya and Gennings, 2024). These different machine learning models are mutually validating and complimentary. Through machine learning approaches, limitations such as linearity, multicollinearity, and proportional hazards when using traditional regression models can be overcome; consequently, given multiple independent variables, a dependent variable can be predicted. We considered a total of 28 lifestyle variables in our machine learning models. In our LASSO, RF and linear model, there were several predictors representing parabens. While, for EDCs: triclosan, BPA, phthalates, and DEHP had fewer selected predictors. Although, there were differences across models, the model performance metrics (RMSE and RMAE) were similar across validation sets. This suggests the presence of key variables related to maternal EDC exposure that may have a dominant influence. Thus, with different machine learning models, cross-validation be performed, and the best-fitting model that show the factors affecting prenatal EDCs were derived using the lowest RMSE and RMAE. While, our findings are limited, the best-fitting machine learning model highlights the important lifestyle factors in predicting prenatal EDCs in pregnant women.

By identifying these crucial factors, their association with prenatal EDCs can be further evaluated. We had different important lifestyle factors affecting early- and late-pregnancy EDC exposure possibly because pregnant women have behavior changes in lifestyle including personal care products, eating and cooking habits, which represent their daily routine, given the influence of environmental and psychological factors (Herzog-Petropaki et al., 2022). Thus, in our study, we calculated the comprehensive lifestyle score of all predicted lifestyle factors derived from machine learning analysis. This comprehensive lifestyle score considered all the lifestyle factors while evaluating their association with prenatal EDC exposure.

To our knowledge, this nationwide study is the first to evaluate the lifestyle factors affecting the prenatal EDC exposure of pregnant women in South Korea. We utilized different machine learning methods to predict the critical lifestyle factors influencing prenatal EDCs. With the

MLM, the complexity and hidden relationships among the lifestyle factors included in the prediction model could be evaluated efficiently. Our study provided a basis for developing risk mitigation strategies regarding prenatal exposures to EDCs through a birth cohort.

This study had some limitations. The sample size of the phthalate database was small because information regarding phthalates was missing among many participants. The characteristics between the included participants and those excluded due to missing information on phthalates were not statistically different (Supplementary Table 3). Considering the study design, we evaluated the indirect association of prenatal EDC exposure by using questionnaire variables. The lifestyle factor data were collected only once at the time of enrollment, which limited our ability to evaluate the association between EDC exposure and lifestyle factors over different time points. This may introduce bias, given the short half-lives of EDCs and the possibility that the recorded lifestyle factors may not fully represent behavior through the entire pregnancy. Similarly, the participants' exposure factors were based on questionnaire responses, which could introduce selection bias. Although we included various sources of EDC exposure in the ML model, we might have missed some sources because of data availability constraints. Moreover, the generalizability of our results was limited because all study participants were Korean women.

5. Conclusions

Essential lifestyle factors such as the use of makeup or perfume affected prenatal Σ Paraben and triclosan exposure. Prenatal Σ DEHP exposure was associated with eating habits (such as consumption of cup noodles) and a high lifestyle score increased prenatal Σ Paraben exposure. Further studies should be performed to develop strategies that can reduce prenatal EDC exposure.

CRedit authorship contribution statement

Surabhi Shah: Conceptualization, Methodology, Visualization, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Jongmin Oh:** Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Yoorim Bang:** Writing – review & editing, Writing – original draft, Investigation. **Seowoo Jung:** Writing – review & editing, Writing – original draft, Methodology. **Hwan-Cheol Kim:** Writing – review & editing, Writing – original draft. **Kyoung Sook Jeong:** Writing – review & editing, Writing – original draft. **Mi Hye Park:** Writing – review & editing, Writing – original draft. **Kyung A Lee:** Writing – review & editing, Writing – original draft. **Jae-Hong Ryoo:** Writing – review & editing, Writing – original draft. **Sanghwan Song:** Writing – review & editing, Writing – original draft. **Huibyeol Park:** Writing – review & editing, Writing – original draft. **Eunhee Ha:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Hwan-Cheol Kim:** Writing – review & editing, Writing – original draft. **Kyoung Sook Jeong:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization.

Data sharing

Data are obtained from the National Institute of Environmental Research (NIER). The authors have no right to share or provide the data.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by a grant (NIER-2019-03-01-001) from the National Institute of Environmental Research (NIER), funded by the Ministry of Environment (MOE), Republic of Korea.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2024.125309>.

Data availability

The authors do not have permission to share data.

References

- Arbuckle, T.E., Davis, K., Marro, L., Fisher, M., Legrand, M., LeBlanc, A., Gaudreau, E., Foster, W.G., Choeurng, V., Fraser, W.D., MIREC Study Group, 2014. Phthalate and bisphenol A exposure among pregnant women in Canada—results from the MIREC study. *Environ Int.* Jul 68, 55–65. <https://doi.org/10.1016/j.envint.2014.02.010>. Epub 2014 Apr 4. PMID: 24709781.
- Braun, J.M., Just, A.C., Williams, P.L., Smith, K.W., Calafat, A.M., Hauser, R., 2014. Personal care product use and urinary phthalate metabolite and paraben concentrations during pregnancy among women from a fertility clinic. *J. Expo. Sci. Environ. Epidemiol.* 24 (5), 459–466. <https://doi.org/10.1038/jes.2013.69>. Epub 2013 Oct 23. PMID: 24149971; PMCID: PMC4016195.
- Chen, H., Quandt, S.A., Grzywacz, J.G., Arcury, T.A., 2011. A distribution-based multiple imputation method for handling bivariate pesticide data with values below the limit of detection. *Environ. Health Perspect.* 119, 351–356. <https://doi.org/10.1289/ehp.1002124>.
- Chen, S., Yu, J., Chamouni, S., Wang, Y., Li, Y., 2024. Integrating machine learning and artificial intelligence in life-course epidemiology: pathways to innovative public health solutions. *BMC Med.* 22 (1), 354. <https://doi.org/10.1186/s12916-024-03566-x>. PMID: 39218895; PMCID: PMC11367811.
- Cho, S., Oh, S., Kim, N.I., Ro, Y.S., Kim, J.S., Park, Y.M., Park, C.W., Lee, W.J., Kim, D.K., Lee, D.W., Lee, S.J., 2017. Knowledge and behavior regarding cosmetics in Koreans visiting dermatology clinics. *Ann Dermatol.* Apr 29 (2), 180–186. <https://doi.org/10.5021/ad.2017.29.2.180>. Epub 2017 Mar 24. PMID: 28392645; PMCID: PMC5383743.
- Ciallella, H.L., Russo, D.P., Aleksunes, L.M., Grimm, F.A., Zhu, H., 2021. Predictive modeling of estrogen receptor agonism, antagonism, and binding activities using machine- and deep-learning approaches. *Lab. Invest.* 101 (4), 490–502. <https://doi.org/10.1038/s41374-020-00477-2>. Epub 2020 Aug 10. PMID: 32778734; PMCID: PMC7873171.
- Di Napoli, I., Tagliaferri, S., Sommella, E., Salvati, E., Porri, D., Raspini, B., Cena, H., Campiglia, P., La Rocca, C., Cerbo, R.M., De Giuseppe, R., 2021. Lifestyle habits and exposure to BPA and phthalates in women of childbearing age from northern Italy: a pilot study. *Int. J. Environ. Res. Publ. Health* 18 (18), 9710. <https://doi.org/10.3390/ijerph18189710>. PMID: 34574636; PMCID: PMC8469822.
- Fruh, V., Preston, E.V., Quinn, M.R., Hacker, M.R., Wylie, B.J., O'Brien, K., Hauser, R., James-Todd, T., Mahalingaiah, S., 2022. Urinary phthalate metabolite concentrations and personal care product use during pregnancy - results of a pilot study. *Sci. Total Environ.* 20 (835), 155439. <https://doi.org/10.1016/j.scitotenv.2022.155439>. Epub 2022 Apr 22. PMID: 35469886; PMCID: PMC11040873.
- Goodrich, J.M., Ingle, M.E., Domino, S.E., Treadwell, M.C., Dolinoy, D.C., Burant, C., Meeker, J.D., Padmanabhan, V., 2019. First trimester maternal exposures to endocrine disrupting chemicals and metals and fetal size in the Michigan Mother-Infant Pairs study. *J. Dev. Orig. Health Dis.* 10 (4), 447–458. <https://doi.org/10.1017/S204017441800106X>. Epub 2019 Jan 30. PMID: 30696509; PMCID: PMC6660406.
- Haggerty, D.K., Upson, K., Pacyga, D.C., Franko, J.E., Braun, J.M., Strakovsky, R.S., 2021. Reproductive toxicology: pregnancy exposure to endocrine disrupting chemicals: implications for women's health. *Reproduction* 162 (5), F169–F180.
- Han, E.J., Choi, K., Sim, S., Choi, J., Uhm, Y., Kim, S., Lim, E., Lee, Y., 2020. Patterns of household and personal care product use by the Korean population: implications for aggregate human exposure and health risk. *Environ. Sci. Eur.* 32, 1–10.
- Herzog-Petropaki, N., Derksen, C., Lippke, S., 2022. Health behaviors and behavior change during pregnancy: theory-based investigation of predictors and interrelations. *Sexes* 3 (3), 351–366.
- Hornung, R.W., Reed, L.D., 1990. Estimation of average concentration in the presence of nondetectable values. *Appl. Occup. Environ. Hyg* 5 (1), 46–51.
- Huang, J.C., Tsai, Y.C., Wu, P.Y., Lien, Y.H., Chien, C.Y., Kuo, C.F., Hung, J.F., Chen, S.C., Kuo, C.H., 2020. Predictive modeling of blood pressure during hemodialysis: a comparison of linear model, random forest, support vector regression, XGBoost, LASSO regression and ensemble method. *Comput. Methods Progr. Biomed.* 195, 105536. <https://doi.org/10.1016/j.cmpb.2020.105536>. Epub 2020 May 22. PMID: 32485511.
- Ingle, M.E., Watkins, D., Rosario, Z., Vélez Vega, C.M., Huerta-Montanez, G., Calafat, A.M., Ospina, M., Cordero, J.F., Alshawabkeh, A., Meeker, J.D., 2019. The association of urinary organophosphate ester metabolites and self-reported personal care and

- household product use among pregnant women in Puerto Rico. *Environ. Res.* 179 (Pt A), 108756. <https://doi.org/10.1016/j.envres.2019.108756>. Epub 2019 Sep 23. PMID: 31574449; PMCID: PMC6905189.
- Jeong, K.S., Kim, S., Kim, W.J., Kim, H.C., Bae, J., Hong, Y.C., Ha, M., Ahn, K., Lee, J.Y., Kim, Y., Ha, E., Ko-CHENS Study group, 2019. Cohort profile: beyond birth cohort study - the Korean Children's Environmental health Study (Ko-CHENS). *Environ. Res.* 172, 358–366. <https://doi.org/10.1016/j.envres.2018.12.009>. Epub 2018 Dec 14. PMID: 30825686.
- Kato, K., Silva, M.J., Reidy, J.A., Hurtz, D. 3rd, Malek, N.A., Needham, L.L., Nakazawa, H., Barr, D.B., Calafat, A.M., 2004. Mono(2-ethyl-5-hydroxyhexyl) phthalate and mono-(2-ethyl-5-oxohexyl) phthalate as biomarkers for human exposure assessment to di-(2-ethylhexyl) phthalate. *Environ. Health Prev. Med.* 112 (3), 327–330. <https://doi.org/10.1289/ehp.6663>. PMID: 14998748; PMCID: PMC1241862.
- Kim, J.H., Kim, D., Moon, S.M., Yang, E.J., 2020. Associations of lifestyle factors with phthalate metabolites, bisphenol A, parabens, and triclosan concentrations in breast milk of Korean mothers. *Chemosphere* 249, 126149. <https://doi.org/10.1016/j.chemosphere.2020.126149>. Epub 2020 Feb 7. PMID: 32062213.
- Luan, H., Zhao, H., Li, J., Zhou, Y., Fang, J., Liu, H., Li, Y., Xia, W., Xu, S., Cai, Z., 2021. Machine Learning for Investigation on Endocrine-Disrupting Chemicals with Gestational Age and Delivery Time in a Longitudinal Cohort. *Research (Wash D C)*, 9873135. <https://doi.org/10.34133/2021/9873135>. PMID: 34755115; PMCID: PMC8548981. (Accessed 18 October 2021).
- Martina, C.A., Weiss, B., Swan, S.H., 2012. Lifestyle behaviors associated with exposures to endocrine disruptors. *Neurotoxicology* 33 (6), 1427–1433. <https://doi.org/10.1016/j.neuro.2012.05.016>. Epub 2012 Jun 26. PMID: 22739065; PMCID: PMC3641683.
- Midya, V., Gennings, C., 2024. Detecting shape-based interactions among environmental chemicals using an ensemble of exposure-mixture regression and interpretable machine learning tools. *Stat Biosci* 16, 395–415. <https://doi.org/10.1007/s12561-023-09405-6>.
- Montrose, L., Padmanabhan, V., Goodrich, J.M., Domino, S.E., Treadwell, M.C., Meeker, J.D., Watkins, D.J., Dolinoy, D.C., 2018. Maternal levels of endocrine disrupting chemicals in the first trimester of pregnancy are associated with infant cord blood DNA methylation. *Epigenetics* 13 (3), 301–309. <https://doi.org/10.1080/15592294.2018.1448680>. Epub 2018 Apr 18. PMID: 29513082; PMCID: PMC5997152.
- Oskar, S., Wolff, M.S., Teitelbaum, S.L., Stingone, J.A., 2021. Identifying environmental exposure profiles associated with timing of menarche: a two-step machine learning approach to examine multiple environmental exposures. *Environ Res.* Apr 195, 110524. <https://doi.org/10.1016/j.envres.2020.110524>. Epub 2020 Nov 26. PMID: 33249040; PMCID: PMC8673778.
- Padmanabhan, V., Moeller, J., Puttabyatappa, M., 2021. Impact of gestational exposure to endocrine disrupting chemicals on pregnancy and birth outcomes. *Adv. Pharmacol.* 92, 279–346. <https://doi.org/10.1016/bs.apha.2021.04.004>. Epub 2021 May 3. PMID: 34452689.
- Park, M., Kim, S., Kim, Y., Nam, D.J., Ryoo, J.H., Lim, S., 2019. Relationship between personal care products usage and triclosan exposure: the second Korean National Environmental Health Survey (KoNEHS 2012-2014). *Ann Occup Environ Med.* Jan 28, 31–32. <https://doi.org/10.1186/s40557-019-0283-y>. PMID: 30705759; PMCID: PMC6348669.
- Qin, J., Yu, T., Li, P., Liu, H., Chen, B., 2019. Using a monotone single-index model to stabilize the propensity score in missing data problems and causal inference. *Stat. Med.* 38 (8), 1442–1458. <https://doi.org/10.1002/sim.8048>. Epub 2018 Nov 22. PMID: 30566258; PMCID: PMC7297448.
- Rocha, B.A., Asimakopoulou, A.G., Honda, M., da Costa, N.L., Barbosa, R.M., Barbosa, F. Jr., Kannan, K., 2018. Advanced data mining approaches in the assessment of urinary concentrations of bisphenols, chlorophenols, parabens and benzophenones in Brazilian children and their association to DNA damage. *Environ Int.* Jul 116, 269–277. <https://doi.org/10.1016/j.envint.2018.04.023>. Epub 2018 Apr 25. PMID: 29704805.
- Serrano, S.E., Braun, J., Trasande, L., Dills, R., Sathyanarayana, S., 2014. Phthalates and diet: a review of the food monitoring and epidemiology data. *Environ. Health* 13 (1), 43. <https://doi.org/10.1186/1476-069X-13-43>. PMID: 24894065; PMCID: PMC4050989.
- Stafoggia, M., Breitner, S., Hampel, R., Basagaña, X., 2017. Statistical approaches to address multi-pollutant mixtures and multiple exposures: the state of the science. *Curr. Environ. Health Rep.* 4, 481–490. <https://doi.org/10.1007/s40572-017-0162-z>.
- Wei, H., Sun, J., Shan, W., Xiao, W., Wang, B., Ma, X., Hu, W., Wang, X., Xia, Y., 2022. Environmental chemical exposure dynamics and machine learning-based prediction of diabetes mellitus. *Sci. Total Environ.* 806 (Pt 2), 150674. <https://doi.org/10.1016/j.scitotenv.2021.150674>. Epub 2021 Sep 29. PMID: 34597539.
- Woodruff, T.J., Zota, A.R., Schwartz, J.M., 2011. Environmental chemicals in pregnant women in the United States: nhanes 2003-2004. *Environ Health Perspect.* Jun 119 (6), 878–885. <https://doi.org/10.1289/ehp.1002727>. Epub 2011 Jan 14. PMID: 21233055; PMCID: PMC3114826.
- Woods, M.M., Lanphear, B.P., Braun, J.M., McCandless, L.C., 2017. Gestational exposure to endocrine disrupting chemicals in relation to infant birth weight: a Bayesian analysis of the HOME Study. *Environ. Health* 16 (1), 115. <https://doi.org/10.1186/s12940-017-0332-3>. PMID: 29078782; PMCID: PMC5658906.
- Wu, X., Zhou, Q., Mu, L., Hu, X., 2022. Machine learning in the identification, prediction and exploration of environmental toxicology: challenges and perspectives. *J Hazard Mater.* Sep 15 (438), 129487. <https://doi.org/10.1016/j.jhazmat.2022.129487>. Epub 2022 Jun 27. PMID: 35816807.